
Daniel Villicana

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Shipped Games

Hamster Ballin'

2026

Game Designer

Created several fully modular systems for designers to use and optimized both CPU and GPU performance.

- Programmed palette swap system that would allow character meshes to change between alternate colors and load this selection data during gameplay
- Aided programmers in writing systems that ranged from single point ball physics, PCG track systems, and kart racing game logic
- Led three teams of level designers, programmers, and artists to create an arcade kart racing game over the course of 20 weeks

Box Shot

Current

Gameplay Programmer

Created several fully modular systems for designers to use and optimized both CPU and GPU performance.

- Programmed base enemy system using C++ in Unreal Engine that could dynamically be derived as blueprints to accomplish different behavior
- Created enemy stacking system that would allow enemies to stack on other enemies and follow their "parent" transform while keeping their original behavior
- Programmed supporting gameplay systems to aide in player experience such as aim assist and first person camera movements

Education

SMU Guildhall,

Masters of Interactive Technology in Digital Game Development

2025 - Current

UC Riverside,

Bachelor of Science, Computer Science

2020 - 2024

Work Experience

Student Aide(Systems Engineering),

Imperial Irrigation District

June 2024 – August 2024

- Worked to automate power outage reports and corrected errors in existing documentation systems.
 - Shadowed systems engineers in order to predict power outages based on power relay data.
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Projects

Doomenstein

Doomenstein: GOLD was a project made at SMU Guildhall using my custom C++ engine. Doomenstein: Gold borrows systems from the original DOOM and Wolfenstein titles in order to create a short boss rush game. State machines are implemented to support boss behavior and a variety of bullet patterns.

NetChess3D

NetChess3D was a project made at SMU Guildhall using my custom C++ engine. NetChess3D uses a custom HLSL Blinnphong shader to implement a rudimentary lighting system as well as normal and sge map rendering. NetChess3D also features peer-to-peer TCP connection to play across different systems.

ButlerHell

ButlerHell was a project made at SMU Guildhall using my custom C++ engine. ButlerHell uses a button and menu system in order to recreate RPG menu navigation to support a combat encounter inspired by the game Undertale. Additionally, the game features a variety of bullet hell attacks controlled by a boss state machine.

Software

- Unreal Engine, Unity Engine
- DirectX11
- Blender, Photoshop,
- .NET, React, AWS

Programming Languages

- C++, C#, Java, Rust, Lua, Rust, UE5 Blueprint
- HLSL, GLSL
- HTML, Javascript, Markdown

Links

Github:

<https://github.com/VillicanaDaniel>

Linkedin:

<http://www.linkedin.com/in/daniel-villicana>
